

G PCM 12

A 12-bit PCM drum machine in the spirit of the big mid-80s digital rhythm programmers.



Overview

G PCM 12 is a drum machine with a sound ROM of 24 drums, built when the plugin loads and stored as 12-bit PCM at 40 kHz - no sample files to install. The set covers punchy and gated kicks and snares, rim, clap, four toms, hats, ride, crash, cowbell, shaker, tambourine, conga, timbale, an electronic tom and a finger snap.

There are 12 pad channels. Each plays one ROM sound, one hit at a time (a new hit restarts the pad), with the classic voice edit: pitch, level, pan, a four-part envelope, a pitch bend that sweeps while the sound plays, reverse, and the MIDI note that plays it. A closed or pedal hat chokes a ringing open hat. The sounds are pitched by resampling and pass through a gentle output low-pass, like the DAC of the original machines.

A pattern sequencer is built in: 8 patterns of 16 steps for all 12 pads, played in time with REAPER's transport, with swing and accents. Patterns can be exported as a MIDI item with a companion script.

Reach for it for

- Gritty 80s digital drum tracks
- Gated snares and big electronic toms
- Programming quick patterns without leaving the plugin
- Reverse and pitch-bent drum effects
- Playing drums from a MIDI keyboard or drum pads

Getting started

1. Insert G PCM 12 on a track (it is an instrument: it has no audio input).
2. To play it from MIDI, arm the track for recording with a MIDI input, or draw notes in a MIDI item. Each pad responds to the MIDI note shown at the right of its row (the TIGHT FUNK kit follows General MIDI drum notes where it can: kick 36, snare 38 and so on).
3. Or use the built-in sequencer: switch PLAY on in the PATTERN section, pick a pattern (A - G hold factory patterns) and start REAPER's transport.
4. Click a pad's name to hear it and select it for VOICE EDIT.

Controls

The header

LCD	Line 1: the selected pad, its sound and the kit. Line 2: the pattern and the sequencer state (SEQ OFF, WAITING FOR PLAY or PLAYING); while playing it shows the step and the tempo.
KITS	Choose a kit: TIGHT FUNK or ELECTRO. A kit sets all 12 pads (sounds, voice edits and notes).
MIDI light	Flashes when a note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The voice matrix

One row per pad. The mini knobs show their value while you touch them; drag them (Shift for fine steps), use the mouse wheel, or double-click to reset.

Pad name	Click to hear the pad and select it for VOICE EDIT. It lights when the pad plays.
Arrow	Opens the list of the 24 ROM sounds for that pad: BD 1, BD 2, BD GATED, SD 1, SD GATED, SD TIGHT, RIM, CLAP, TOM 1 - 4, HH CLOSED, HH OPEN, HH PEDAL, RIDE, CRASH, COWBELL, SHAKER, TAMBOURINE, CONGA, TIMBALE, ELEC TOM and SNAP.
Steps 1 - 16	The pad's steps in the current pattern. Click to cycle off, hit, accent (accents are brighter, with a white bar). The playhead is shown while the sequencer runs.
PITCH	-24 to +24 semitones. Pitching is done by resampling, so higher is also shorter and lower is longer.
LEVEL	0 to 100 (80 by default).
PAN	-100 (left) to +100 (right).
DECAY	The pad's DECAY 1 time, 10 ms to 4000 ms; fully up leaves the sound as stored.
BEND	-100 to +100: the pitch sweeps down or up while the sound plays (up to 4 octaves per second at full).
REV	Plays the sound backwards.
NOTE	The MIDI note that plays the pad, 24 to 96. If two pads share a note, the one nearer the top of the list plays.

VOICE EDIT (the selected pad)

The envelope shapes the stored sound: an attack ramp, DECAY 1 towards the decay point, then DECAY 2 to silence. Decay times are the time to fall by 40 dB.

ATTACK	A fade-in at the start of the sound, 0 to 200 ms.
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DECAY 1	10 to 4000 ms; fully up reads FULL and leaves the sound's own decay untouched.
D.POINT	The level DECAY 1 falls to before DECAY 2 takes over, 0 to 100%. At 100% DECAY 1 has nothing to do and DECAY 2 shapes the whole sound.
DECAY 2	10 to 4000 ms; FULL leaves the rest of the sound as it is. A short DECAY 2 after a high D.POINT gives a gated, chopped tail.
BEND	As on the pad's row: DOWN or UP and the amount, OFF at the centre.
REVERSE	As REV on the pad's row.

PATTERN

PLAY	Switches the sequencer on. It then plays whenever REAPER's transport runs, locked to the project's beat position (it counts four beats to the bar).
A - H	Choose the pattern. While playing, a new pattern starts at the next bar (its key blinks until then); when stopped it changes at once.
SWING	0 to 75%: delays every second 16th.
COPY TO	Copies the current pattern to another slot (A - H).
CLEAR	Empties the current pattern.
BARS (1 / 2 / 4 / 8)	How many bars EXPORT writes (the pattern is repeated).
EXPORT	Hands the current pattern to the export script (see below). The caption under it reads 'run script' if the script did not pick the pattern up.

Accented steps play at full velocity, normal hits at 70%. The patterns and the pad set are saved with the project.

MASTER

VOLUME	Output level, -24 to +12 dB.
VELOCITY	How much MIDI velocity (and accents) changes the level, 0 to 100% (60% by default). At 0 every hit plays at the same level.
DAC	The output low-pass filter of the converter, 4 to 20 kHz (14 kHz by default). Lower it for a duller, older sound.
CH	The MIDI channel the pads listen to: OMNI (all) or 1 to 16.
Meters	Output level, left and right. Click to reset the peak hold.

Kits

TIGHT FUNK	Short, dry and snappy (the default). Pads: BD 1, SD TIGHT, SD 1, RIM, CLAP, TOM 1, TOM 2, TOM 3, HH CLOSED, HH OPEN, HH PEDAL, COWBELL. The factory patterns are written for this kit.
ELECTRO	Electronic toms raised and shortened with a gentle downward bend, short sounds, a bent kick and BD GATED on the last pad.

Patterns

A 80S ROCK	Kick, backbeat snare, eighth-note closed hats and an open-hat accent.
B FUNK	Syncopated kick and tight snare with ghost notes, busy 16th hats.
C ELECTRO	Electro kick with claps, eighth hats and cowbell.
D DISCO	Four-on-the-floor kick, backbeat and off-beat open hats.
E GATED BALLAD	A slow backbeat with eighth hats and a rim.

F TOM FILL	A tom fill down through the three toms.
G CLAPS + BELL	Four-on-the-floor with claps, a cowbell line and off-beat hats.
H USER	Empty, ready for your own pattern.

Exporting patterns as MIDI

JSFX plugins cannot write files, so EXPORT works with a companion ReaScript, 'GFX G-PCM 12 - export MIDI.lua' in the Scripts folder.

1. In REAPER, open Actions > Show action list, choose New action > Load ReaScript, and pick 'GFX G-PCM 12 - export MIDI.lua'. Select it and press Run.
2. The script keeps running in the background (its toolbar button lights if you add it to a toolbar) and exports every time you press EXPORT. Run it again to stop it.
3. Select the track you want the MIDI on and place the edit cursor where the pattern should start.
4. Choose BARS (1, 2, 4 or 8) and press EXPORT. A MIDI item named 'G-PCM 12 pattern' appears at the edit cursor.
5. If you pressed EXPORT before starting the script, it exports that pattern as soon as it starts.

Each pad's row is written on that pad's MIDI note, on MIDI channel 10; hits have velocity 100 and accents 127, and the swing is kept.

Tips

- For the classic 80s snare, use SD GATED, or set a high D.POINT and a short DECAY 2 on SD 1 to chop its tail.
- Pitching a sound down makes it longer and grainier - the 12-bit character comes through more.
- A low BEND on the toms or kick gives the electronic 'pew' of the era; ELECTRO starts you there.
- Use COPY TO to make variations of a pattern, then switch between them while playing - the change waits for the bar line.

File: gfx-g-pcm12.jsfx